We can do the same thing with $\mathbf{a}^+$ and **N**. The result is almost the same except for the sign. Here is the whole list of commutators in one neat package:

$$\begin{aligned} [\mathbf{a}^-, \mathbf{a}^+] &= 1 \\ [\mathbf{a}^-, \mathbf{N}] &= \mathbf{a}^- \\ [\mathbf{a}^+, \mathbf{N}] &= -\mathbf{a}^+. \end{aligned} \qquad (10.25)$$

This is what you might call a commutator algebra: a set of operators that *closes* under commutation. Commutator algebras have wonderful properties that make them one of the theoretical physicist's favorite tools. We are now going to see the power of this commutator algebra in the iconic example of the harmonic oscillator, using it to find the eigenvalues and eigenvectors of **N**. Once we know these, we can immediately read off the eigenvalues of **H** from Eq. 10.23. The trick is to use a kind of induction procedure: we begin by supposing we have an eigenvalue and eigenvector of **N**. Call the eigenvalue *n* and the eigenvector |*n*>. By definition,

$$\mathbf{N}|n\rangle = n|n\rangle.$$

Now, let's consider a new vector, obtained by acting on |*n*> with $\mathbf{a}^+$. Let's prove that the result is a different eigenvector of **N**, with a different eigenvalue. Again, we accomplish this by straightforward application of the commutation relations. We'll start by writing the expression $\mathbf{N}(\mathbf{a}^+|n\rangle)$ in a slightly more complicated form,

$$\mathbf{N}(\mathbf{a}^+|n\rangle) = [\mathbf{a}^+\mathbf{N} - (\mathbf{a}^+\mathbf{N} - \mathbf{N}\mathbf{a}^+)]|n\rangle.$$

The expression in brackets on the right-hand side is the same as $\mathbf{N}\mathbf{a}^+$, with the term $\mathbf{a}^+\,\mathbf{N}$ added and then subtracted. But notice that the expression in parentheses is the last of the commutators from Eqs. 10.25. If we plug that in, we get